

Comparative Efficacy of ICSI Versus Conventional IVF in Non-Male Factor Infertility

Kenisya Saravenen | BSc Biomedical Sciences | University of York

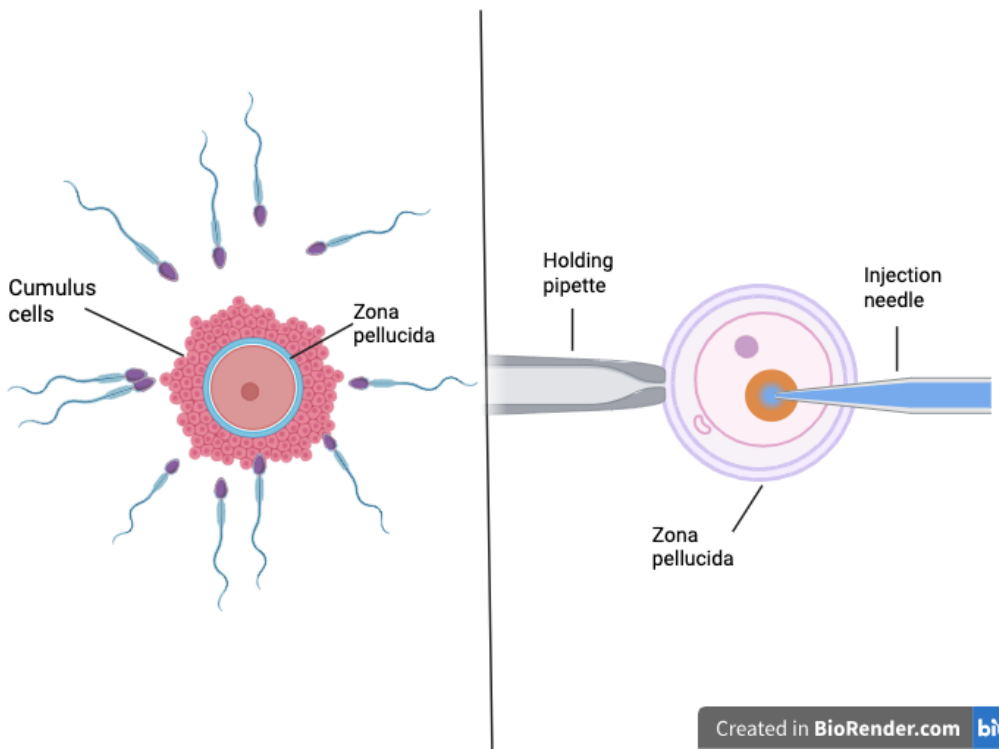
LITERATURE-BASED EVIDENCE SYNTHESIS • 2026

KEY QUESTION — When severe male-factor infertility is absent, does bypassing natural fertilisation improve outcomes that matter to patients?

01 BACKGROUND & BIOLOGICAL MECHANISMS

ASSISTED REPRODUCTIVE TECHNOLOGY

Conventional IVF (cIVF) and intracytoplasmic sperm injection (ICSI) both fertilise oocytes outside the body, but they expose spermatozoa to very different biological selection pressures. In cIVF, sperm must interact with the cumulus–oocyte complex, reach the oocyte, bind to the zona pellucida and undergo the acrosome reaction before membrane fusion. ICSI instead places a single selected sperm directly into the oocyte cytoplasm, bypassing several of these steps. [5,6]



Why cIVF is biologically distinct

cIVF retains a sequence of functional tests: sperm motility, migration, zona pellucida interaction, the acrosome reaction and oolemma fusion. These processes do not guarantee a competent embryo, but they do mean that sperm must complete several stages before fertilisation can occur. [5,6]

What ICSI bypasses

ICSI can overcome a failure of sperm–oocyte interaction because the embryologist immobilises one spermatozoon and injects it directly into the oocyte. Oehninger and Gosden argued that this is particularly relevant when semen testing or previous fertilisation history indicates impaired fertilising capacity. [6,7]

THE UNRESOLVED BIOLOGICAL QUESTION

The trade-off is that ICSI solves one laboratory problem by removing several natural selection steps. Modern sperm-selection methods attempt to recover some of this selection before injection. The 2021 review describes approaches including density-gradient centrifugation, swim-up, membrane-based selection, high-magnification assessment and microfluidics, but concludes that uncertainty remains over which technique provides the most clinically meaningful advantage. [5]

Mechanism → outcome

A key principle for interpreting ART studies is that a laboratory endpoint is not equivalent to a patient endpoint. Increased fertilisation can be useful, but the clinically important outcomes are embryo development, pregnancy and ultimately live birth. The literature therefore needs to be judged across the entire pathway rather than at fertilisation alone.

02 CLINICAL INDICATIONS & THE SHIFT IN PRACTICE

WHERE ICSI HAS A CLEAR BIOLOGICAL RATIONALE

ICSI was introduced for couples in whom conventional fertilisation was difficult because of severe male-factor infertility. Oehninger and Gosden emphasised that when semen analysis and sperm-function testing indicate impaired fertilising capacity, couples may be directed towards ICSI; they also highlighted previous fertilisation failure as a setting in which ICSI can improve the chance that an egg will fertilise. [6] The 2024 Lancet trial likewise notes that ICSI was originally introduced for severe male infertility before its use expanded to less severe indications. [1]

SEVERE MALE-FACTOR INFERTILITY

The strongest indication is a major impairment in sperm quantity or function, where direct injection can bypass barriers that conventional IVF depends upon. The mechanistic rationale is therefore different from using ICSI simply because a fertilisation attempt is taking place.

PREVIOUS POOR FERTILISATION

ICSI may be considered after previous IVF has produced poor or failed fertilisation, because the direct injection approach can bypass a suspected sperm–oocyte interaction problem. Oehninger and Gosden described this as potentially revealing an otherwise ‘occult’ gamete defect. [6]

WHY USE HAS EXPANDED

ICSI is now frequently used outside severe male-factor infertility. The 2026 meta-analysis describes this as a widespread practice despite insufficient evidence for benefit in couples without severe male factor. One proposed reason is that clinics may prioritise avoiding total fertilisation failure, particularly where only one treatment cycle is feasible. [3,4]

THE FERTILITY-PRESERVATION DILEMMA

Aggarwal et al. examined IVF versus ICSI for fertility preservation and found that ICSI produced more fertilised eggs and embryos for storage or later treatment. However, the authors reported that 19% of eggs were not used in ICSI treatment and concluded that the higher fertilisation yield did not automatically make ICSI the better overall option. They argued that clinics should select IVF or ICSI according to sperm characteristics rather than use ICSI for every fertility-preservation cycle. [4]

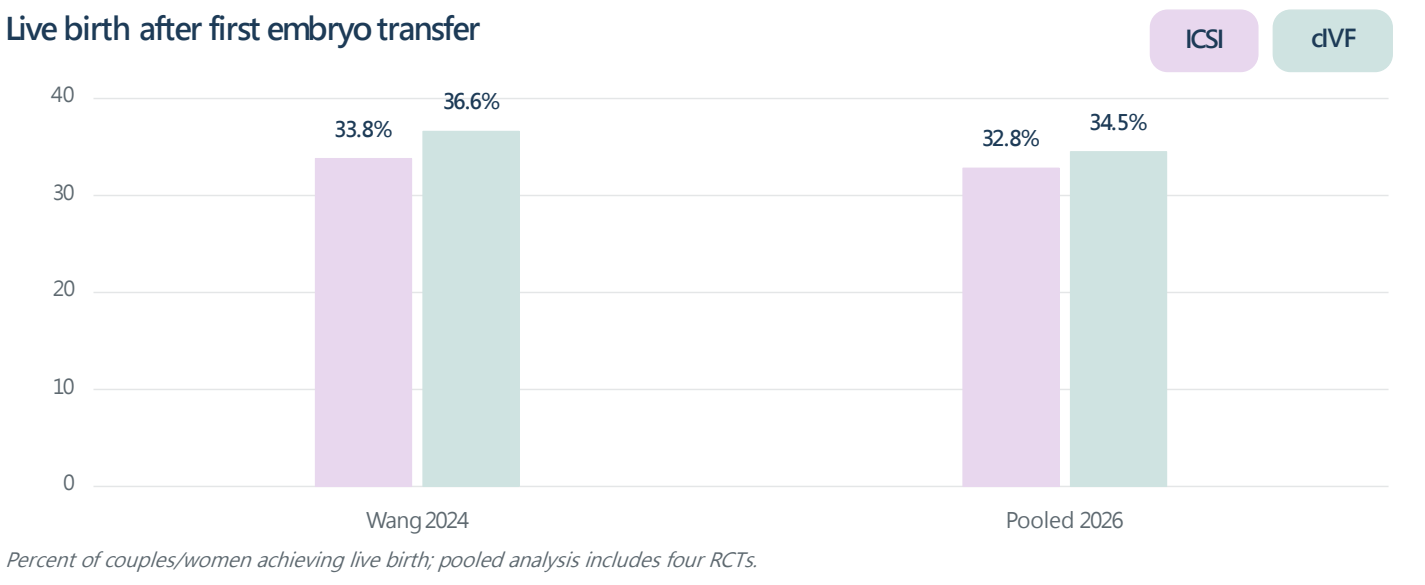
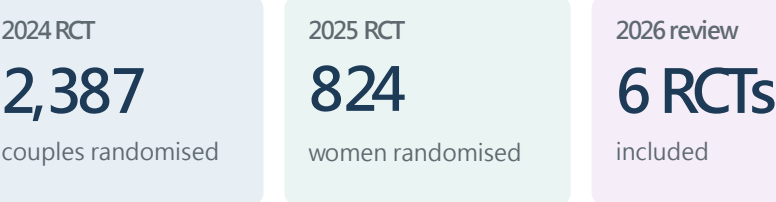
A useful clinical distinction

The question is not whether ICSI works—it clearly can—but whether its additional invasiveness, laboratory time and cost are justified when sperm quality is not severely impaired and conventional IVF already provides a reasonable route to fertilisation.

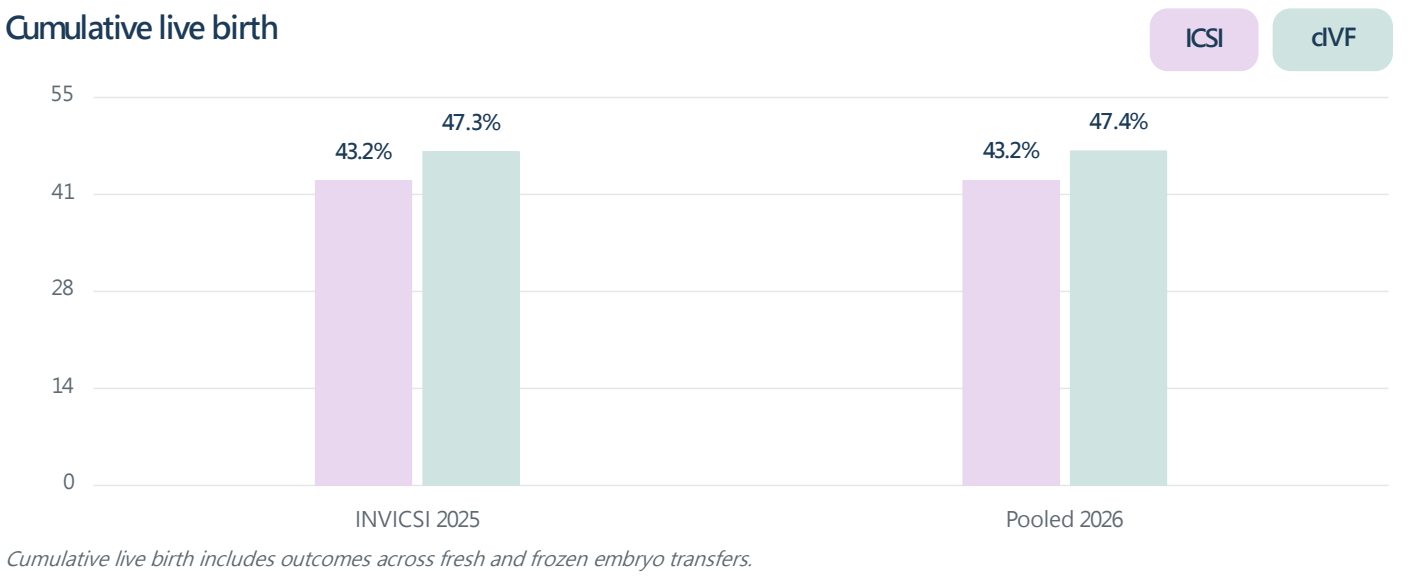
03 CLINICAL EVIDENCE, OUTCOMES & STATISTICS

THE EVIDENCE BASE

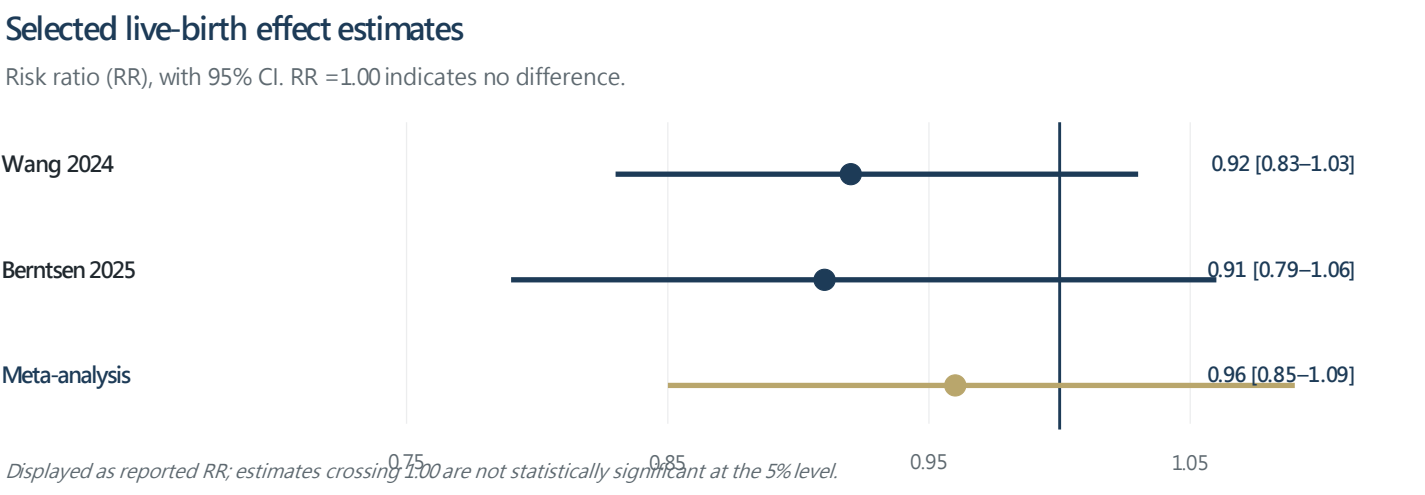
The strongest evidence in this poster comes from randomised controlled trials and a 2026 systematic review/meta-analysis. Wang et al. randomised 2,387 couples across ten Chinese reproductive-medicine centres. Berntsen et al. randomised 824 women across six Danish public fertility clinics. Kayimu et al. then identified six trustworthy RCTs in couples without severe male-factor infertility and assessed risk of bias using RoB 2, study trustworthiness using TRACT and certainty with GRADE. [1–3]



In the 2024 Lancet trial, live birth after the first embryo transfer occurred in 33.8% of couples assigned to ICSI and 36.6% assigned to cIVF (adjusted RR 0.92, 95% CI 0.83–1.03; p=0.16). The result did not demonstrate an improvement with ICSI. [1] The 2026 meta-analysis similarly found 32.8% vs 34.5% across four RCTs, with a pooled RR of 0.96 (95% CI 0.85–1.09), rated as high-quality evidence. [3]



The INVICSI trial reported cumulative live birth rates of 43.2% with ICSI and 47.3% with cIVF (RR 0.91, 95% CI 0.79–1.06). [2] The 2026 meta-analysis pooled three RCTs and found 43.2% versus 47.4% (RR 0.92, 95% CI 0.84–1.01; high-quality evidence). [3] These confidence intervals cross 1.00, so the estimates do not provide a statistically significant advantage for ICSI.



The same meta-analysis found no significant difference in clinical pregnancy (37.7% vs 39.4%; RR 0.96, 95% CI 0.88–1.04) or fertilisation rate (mean difference –0.01, 95% CI –0.01 to 0.03), although heterogeneity for fertilisation rate was high (I²=87%). Total fertilisation failure was also comparable (4.3% vs 5.0%; RR 0.86, 95% CI 0.65–1.13). [3]

04 INTERPRETATION, SAFETY, LIMITATIONS & CONCLUSION

WHAT THE RESULTS MEAN BIOLOGICALLY

The evidence does not show that bypassing natural fertilisation steps produces more live births in couples without severe male-factor infertility. This is important because ICSI can change the rate of fertilisation without necessarily changing downstream embryo competence or the probability of live birth. The 2025 secondary analysis of INVICSI also reported fewer usable and high-quality day-5 blastocysts after ICSI than cIVF, while observing no difference in early developmental kinetics. [2,8]

SAFETY: WHAT CAN BE SAID WITH CONFIDENCE?

ICSI is invasive and adds a laboratory micromanipulation step. The 2024 RCT authors highlighted additional cost and potential offspring-health risks as reasons not to use ICSI routinely without a clear indication. [1] The 2026 meta-analysis did not identify significant differences in birth-defect rate, low birth weight, miscarriage or most other pregnancy outcomes, but several safety outcomes were based on relatively small numbers. [3] This supports cautious interpretation rather than a claim that the two techniques are biologically identical.

THE SPERM-SELECTION PROBLEM

Liu and Baker emphasised that semen assessment is important both for diagnosing male infertility and for selecting treatment. [7] This connects directly to the 2021 review by Baldini et al., which examined methods designed to enrich the sperm population used for ICSI. Swim-up, density-gradient centrifugation and microfluidic systems can select sperm according to different functional characteristics, but the review notes ongoing disagreement about which method should be considered the ‘clinical winner’. [5]

LIMITATIONS OF THE EVIDENCE

The 2026 meta-analysis included only six RCTs meeting its predefined trustworthiness criteria. The authors cautioned that live-birth data came from a limited number of high-quality studies, subgroup evidence was limited and outcome definitions varied. Fertilisation rate also showed substantial statistical heterogeneity. [3] Results therefore support evidence-based restraint, not the claim that every clinical subgroup has been definitively resolved.

CONCLUSION

For couples without severe male-factor infertility, current randomised evidence does not show a live-birth advantage for routine ICSI over conventional IVF. ICSI remains biologically and clinically important where severe male-factor infertility or previous fertilisation failure creates a specific rationale for bypassing conventional fertilisation. The central lesson is therefore not ‘ICSI versus IVF’ in isolation, but matching the fertilisation method to the underlying biological problem and judging success by patient-centred outcomes.

SELECTED REFERENCES

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